

When Agent Failure Prediction Measures Task Difficulty Rather Than the Run

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Predictors of LLM-agent failure report AUROC between 0.85 and 0.94, and these figures are read as evidence that a running agent can be monitored and stopped early when it goes wrong. We show that the metric cannot support that reading. Pooled AUROC decomposes exactly into a same-task and a cross-task component, and the same-task component—the only one that measures whether a predictor can tell two attempts at one task apart—carries weight $\Theta(1/n)$: between 0.003% and 0.07% on our corpora. A pooled figure in the published band is therefore equally consistent with a genuine run-level monitor and with a predictor that only estimates task difficulty, whose ceiling is computable from the outcome table alone (0.95–0.99 on our corpora). The run-level question has to be measured directly.

We measure it on four corpora that record repeated attempts at the same task by the same model (153,944 trajectories from 21 models, plus a 9,989-run activation-probe corpus). A task-difficulty oracle that never observes the run reaches pooled AUROC 0.894; annotated task properties alone reach 0.826; adding the entire trajectory to the oracle adds 0.029. Within task, early discrimination is near chance: 0.51 after one turn for a reimplemented published monitor, and 0.49–0.52 for a second system’s *released* pipeline run unmodified across an entire public corpus, which nonetheless reaches pooled 0.72 *before the agent acts*. On a corpus of weaker acting models a weak early signal appears (0.57 by five steps), the direction predicted by a capability trend we identify. In replay against a task-level allocation policy, a monitor would need within-task AUROC of roughly 0.84–0.93 before aborting adds throughput, and a perfect monitor adds at most 7.8 tasks. Runtime failure monitors should therefore be evaluated within task, on prefixes a deployed monitor could compute, and against task-level allocation baselines before being used for intervention; task-level prediction remains useful for triage, routing and benchmark efficiency.

CCS Concepts: • **Software and its engineering** → **Software verification and validation**; *Software testing and debugging*; • **Computing methodologies** → *Machine learning*.

Additional Key Words and Phrases: LLM agents, failure prediction, evaluation validity, measurement, empirical software engineering

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1 INTRODUCTION

LLM agents now do multi-step software-engineering work, and they fail often: measured failure rates run from 41% to 86.7% across seven frameworks [7], and each failure costs minutes and many tokens. A growing literature therefore reads the partial execution trace of a running agent and scores impending failure, reporting strong numbers: verbal-confidence AUROC 0.85 at completion [23], hidden-state probes reported to anticipate failure from the first round on TextCraft with small open models while saving 54–60% of tokens [31], and automaton monitors reaching 0.941 [9], across ICML, NeurIPS, ICLR, EMNLP and arXiv.

Three anomalies. Three published findings sit awkwardly beside those numbers.

First, accuracy does not translate into deployment benefit. A critic with offline AUROC 0.94, used to intervene in running agents, made end-to-end performance *worse*, disrupting up to 7.3 successes per recovered failure [39]. **Second**, a model that never observes the agent does about

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as well. Predicting success *before the run starts*, from static task properties alone, reaches AUROC 0.841 on held-out SWE-bench Verified [15, 19], inside the range reported by systems that read entire trajectories; task-difficulty prediction and trajectory monitoring have largely developed as separate literatures. **Third**, much of the outcome is not a property of the task at all. The same agent on the same task yields one success and one failure roughly a fifth of the time [22], and at the moment such a pair begins, the two runs are identical in every observable respect.

Our hypothesis. Consider a forecaster evaluated across Seattle and Phoenix. “Rain in Seattle, sun in Phoenix” scores well while knowing nothing about weather, and asked whether it will rain in Seattle *tomorrow*, it offers nothing. The reported metric has this structure: it pools runs from many tasks and asks whether failures rank above successes overall. Failures come disproportionately from hard tasks, so a predictor that estimates only task difficulty scores well having learned nothing about any individual run, and run-level discrimination is precisely what aborting requires. In short:

A predictor that cannot separate two attempts at the same task knows only the task. It can justify moving effort between tasks; it cannot justify aborting the attempt in front of it on that attempt’s evidence.

Software engineering has met both halves of this problem. The pooled-versus-within distinction is the agent-monitoring form of cross-project versus within-project defect prediction [52]; the lesson that a predictive metric must match the decision it informs is likewise SE-native [30], as is the concern that accuracy misleads on the imbalanced data typical of defect prediction [12, 27]. Mixed-outcome units are the agent-run analogue of flaky tests [25]: our 22–41% observed mixed rates, under each corpus’s available repeat count, play the role that flakiness rates play for tests, and a flaky unit’s outcome can no more be read off the task’s identity than a flaky test’s verdict can be read off the test’s.

1.1 What is already known, and what is not

Four layers of this argument have different status. It is *known* that pooled ROC can differ from covariate-specific discrimination; this is biostatistics’ covariate-adjusted ROC [18, 29]. Within-task controls are *beginning to appear*: Mehtiyev and Assunção [26] show that a coding agent’s trajectory-length signal reverses inside a task, and Bulut [6] separates budget and difficulty confounds in reasoning traces with per-problem oracles. Neither carries the pair decomposition, the $\Theta(1/n)$ collapse, or multi-corpus within-task measurement of failure *predictors*. Our *diagnosis* is that in repeated agent benchmarks the same-task pair weight collapses as $\Theta(1/n)$, so failure-prediction evaluations, published monitors included, are dominated by cross-task comparisons. Our *consequence* is that a strong pooled AUROC can mean weak support for the decision the monitor exists to make, and that selecting policies by it selects worse ones.

Two recent studies already report weak early failure-prediction signals in their settings: Li et al. [23] via path switching at 50% progress, and Lee [22] via canonical-path deviation on mixed-outcome units, whose identification strategy we adopt directly (§3.2). We claim no novelty there. Our claims are scoped to prefixes, the regime in which intervention is possible, and the open question is metrological: what do the high reported accuracies measure? Neither study decomposes the reported accuracy. That decomposition is what this paper supplies, together with its consequences: a diagnostic, a reproduction on the strongest available features, and a replacement policy that outperforms the recommendation Li et al. [23] derive from their own findings.

Some benchmarks record several independent attempts at the same task by the same model. Within such a unit, task and model are fixed by construction and only the run varies, so measuring accuracy *inside* units isolates run-level discrimination, and comparing it with the pooled figure separates the questions the pooled metric conflates.

1.2 Research questions

- RQ1** (*Estimand*) What does pooled failure-prediction accuracy measure in benchmarks that record repeated attempts?
- RQ2** (*Run-level signal*) How much run-specific information is available at online-computable prefixes, across predictors, models, corpora and hidden-state probes?
- RQ3** (*Deployment*) Does evaluating the correct estimand change intervention and resource-allocation conclusions?

Two scoping notes. The allocation policy of §6 presupposes prior attempts at the task by some model and targets repeated-evaluation settings (benchmark runs, CI re-execution, model migration) rather than a first attempt at a novel issue. Within RQ2, the hidden-state probe is the strongest objection to a behavioural null—a reader may grant that visible behaviour carries no early run-level signal yet hold that it is latent in the acting model’s activations, the premise of an active probing literature—so we test it on the model that produced the trajectories.

1.3 Contributions

- **Estimand analysis (RQ1).** An exact pairwise decomposition $A_{\text{pool}} = w A_{\text{within}} + (1 - w) A_{\text{cross}}$ whose same-task weight is $w = \Theta(1/n)$. On our corpora w lies between 3.4×10^{-5} (C2) and 6.8×10^{-4} (C4-L), so at least 99.93% of the pairs behind a pooled figure compare different units. The pooled-versus-conditional distinction is established statistics [18, 29]; its extreme, load-bearing form in agent evaluation is the contribution, together with a difficulty-only ceiling computable from a benchmark’s outcome table (§2.3).
- **Empirical diagnosis (RQ2).** Across four corpora, 21 models and three scaffolds (153,944 trajectories in C1–C3, plus a 9,989-run probe corpus from two further models), a task-difficulty oracle that never observes the run reaches pooled 0.894, deployable task metadata alone reaches 0.826, the full trajectory adds +0.029, and early within-task discrimination sits at or near our resolution boundary, for behavioural predictors and for linear probes of the acting model’s hidden state alike (§5.1–§5.3).
- **Validation on published protocols (RQ2).** A reimplemented published monitor reproduces its reported SWE-agent pooled AUROC (0.806 vs 0.799) yet is at chance within task at one turn (0.507), and its fractional checkpoints leak eventual trajectory length worth +0.064 (§5.4). A second system’s *released* pipeline, run unmodified on five pre-registered folds covering every one of C2’s 6,306 instances, reaches pooled 0.72 with no run observed, stays at chance through five observed steps and never exceeds 0.58 within task by run end; on C1’s weaker acting models the same pipeline finds a weak early signal (0.57 by five steps), so the early null is a property of strong acting models rather than of agents generally (§5.5).
- **Deployment consequences (RQ3).** In budgeted replay with abort thresholds tuned on held-out tasks (Table 3), working tasks in a *static* difficulty order transferred from another model solves up to 3.5× more issues than round-robin (adaptive sampling shifts this by −6 to +6%), a tuned abort-only policy solves 11–68% fewer than either, and adding the tuned monitor to allocation changes throughput by −4 to −1 tasks with every 95% CI spanning zero. A pre-registered sweep places the within-task AUROC a monitor would need at ≈ 0.84 (10M) to ≈ 0.93 (5M), with a perfect monitor adding at most +7.8 tasks (§6.3). Scripts, tests, pre-registrations and every table are released identity-free (§8).

The conditions are the result. Our findings come with conditions: early run-level signal is absent on strong acting models and weak on weaker ones, and the discrimination a monitor would need is 0.84–0.93 on one corpus and out of reach even at perfection on another. They are easy to read as

hedging; here they are the finding. A pooled score does not transfer across models, scaffolds and corpora; that is the claim, and if our own numbers transferred unconditionally the paper would be refuting itself. Two measurable quantities govern the variation. Acting-model capability sets how legible a run is ($\rho = -0.66$ across 17 models, -0.75 within families, and reproduced on a released system’s own predictions; §5.6). The strength of the policy a monitor is compared against sets how much discrimination must be bought before aborting pays (§6.3). Hold those two fixed and the measurements are stable; leave them unstated, as pooled evaluation does, and no reported number constrains what happens in the next setting.

2 ESTIMANDS: POOLED AND WITHIN-TASK DISCRIMINATION

This section defines every quantity we use before we use it. Readers familiar with AUROC may still want §2.2, which introduces the distinction that organises the paper.

Runs, units, and mixed-outcome units. A *run* is one complete attempt at a task, labelled success or failure by the benchmark’s oracle. A *unit* is a (model, task) pair; repeated runs of one unit hold task and model fixed, so only the run varies. A unit is *mixed* when it has at least one success and one failure; only mixed units carry information about A_{within} , a locality we analyse in §7. AUROC has the pairwise reading we use throughout: pick one failed and one successful run at random; AUROC is the probability the predictor scores the failed run higher (ties count half). 0.5 is a coin flip.

2.1 Three questions, three estimands

One number is routinely treated as the answer to three different questions.

Q1: cross-task triage (*which tasks will fail?*): task-level prediction; pre-execution difficulty estimation (§6.2) serves it. **Q2: run-level monitoring** (*given this task, which attempt will fail?*): the question a monitor answers when it aborts the run in front of it; within-task discrimination is its estimand, and it is this paper’s subject. **Q3: benchmark-wide ranking** (*can a score rank all successes above all failures?*): pooled AUROC answers this faithfully—the literature routinely answers Q3 while motivating the work by Q2.

Within-task AUROC isolates the *incremental* run-specific information once task and model are fixed—the diagnostic for the claim that the current trace distinguishes which attempt will fail. It is not the complete decision metric: AUROC measures discrimination, not calibration or intervention utility, and a task-level score at within-task 0.5 can still serve triage (Q1), as our allocation results show; §6 evaluates actual policies under solved-under-budget and break-even objectives.

2.2 Pooled versus within-task AUROC

The distinction that organises this paper is *which pairs* of runs the probability above ranges over. **Pooled AUROC** (A_{pool}) draws the failed and successful run from anywhere in the dataset—usually different tasks; it is the number the literature reports. **Within-task AUROC** (A_{within}) draws both from the *same unit*, computed per unit and pair-weighted:

$$A_{\text{within}} = \frac{\sum_u n_u A_u}{\sum_u n_u}, \quad n_u = |S_u| \cdot |F_u|, \quad (1)$$

where S_u and F_u are the successful and failed runs in unit u , and A_u is the AUROC computed within that unit.

A score that says only “this task is hard” is the same on every run in the unit, so it scores exactly 0.5 on A_{within} ; anything above 0.5 must reflect the individual run. We call $A_{\text{pool}} - A_{\text{within}}$ the **gap**, a diagnostic of between-task inflation. It is deliberately not an *attribution*, since the two are different pairwise estimands; what decomposes exactly is A_{pool} itself, which we turn to next.

2.3 An exact decomposition of pooled AUROC

Pooled AUROC is a probability over positive–negative pairs, which partition into same-unit and cross-unit pairs. Writing w for the share of pairs that are same-unit, this partition gives an identity:

$$A_{\text{pool}} = w A_{\text{within}} + (1 - w) A_{\text{cross}}, \quad w = \frac{\sum_u P_u N_u}{(\sum_u P_u)(\sum_u N_u)},$$

where P_u and N_u count the failing and succeeding runs of unit u , and A_{cross} is concordance over cross-unit pairs. This holds by construction for any scoring function; we verify it numerically to 10^{-9} in every configuration we report.

The identity matters because of how small w is. For n units of r runs each, the same-unit pairs number $O(nr^2)$ against $O(n^2r^2)$ cross-unit pairs, so $w = \Theta(1/n)$ as a benchmark adds units with runs per unit bounded—the regime of every corpus here, where benchmarks grow by adding tasks, not repeats. On C4-L (371 units, 142 mixed), $w = 0.0007$: 2,259 within-unit pairs against 3,314,835 cross-unit—99.93% of the positive–negative pairs behind pooled AUROC compare *across* tasks, so the A_{within} term is invisible in the pooled number. Two readings must then be kept apart. The decomposition partitions *pairs*, not *information*: a feature that reads the run also helps rank a failed run of one task above a successful run of another, and so enters A_{cross} . $\Theta(1/n)$ is therefore an *identifiability* statement—a pooled figure cannot distinguish a predictor that reads the run from one that only estimates task difficulty—not, by itself, evidence that run-level signal is absent. Whether it is present must be measured directly, which A_{within} and the nested $M_0/M_1/M_2$ comparison (§5.2) do.

The identity also yields a *predictive* theory of the confound. With μ the mean failure rate and MD the Gini mean difference of the unit rates q_u , the difficulty oracle’s pooled AUROC has closed form $\frac{1}{2} + \text{MD}/(4\mu(1 - \mu))$ and the same-unit weight the constant $w = \overline{q}(1 - \overline{q})/(n\mu(1 - \mu))$ —both computable from a benchmark’s *outcome table alone* (drawn runs weighted by unit size, i.e. pairs sampled as in the AUROC itself; derivation in the supplement). The ceiling predicts 0.954/0.987/0.984/0.973/0.979 for our five corpora/splits against freshly measured leave-one-run-out oracles of 0.945/0.974/0.903/0.949/0.959; and the attenuation is itself derivable—a drawn run’s leave-one-out score is $\text{Bin}(r-1, q_u)/(r-1)$ exactly, giving a finite-run formula that roughly halves the ceiling’s error on every corpus (0.950/0.980/0.940/0.962/0.969; residual falls with runs per unit, from 0.037 at C3’s three to 0.005–0.013 at ten or more, its sign and size consistent with plug-in dispersion bias). Derivations in the supplement; the w predictions match §4.1 exactly. Larger benchmarks make the mismatch worse: w falls as $1/n$ while the ceiling depends only on difficulty spread. Formally, A_{within} is a member of the covariate-adjusted ROC family [18] with covariate = unit; it is not identical to their estimator—they weight covariate-specific AUCs by the case distribution, we by pair counts $P_u N_u$ —and the agent-specific content here is the $\Theta(1/n)$ collapse, the ceiling formula, and their deployment consequences, not the family.

2.4 Variance decomposition and deployment accounting

The intraclass correlation (ICC) [28] is the share of outcome variance lying *between* units. Its naive estimator $\text{Var}(\bar{y}_u)/\text{Var}(y)$ is upward-biased when runs per unit are few, so we report a finite-repeat-corrected value (§5.2), and we use it only as descriptive evidence of heterogeneity, not as a bound on what a run-level predictor can explain. For the deployment analysis (RQ3) we adopt the accounting of Vasudev et al. [39]: with recovery rate r (restart succeeds given the original would fail) and disruption rate d (restart fails given the original would succeed), intervening pays only when the baseline failure rate exceeds the break-even $d/(r + d)$.

2.5 Uncertainty and hypothesis tests

Confidence intervals. All intervals are 95% percentile bootstrap intervals resampling *units*, not runs (2000–5000 resamples); resampling runs would understate uncertainty, since runs inside a unit are correlated by construction.

Early-prefix hypothesis tests. Where we ask whether useful run-specific discrimination is present at intervention-relevant prefixes we test $H_0 : A_{\text{within}} = 0.5$ with a within-unit conditional permutation test on held-out cross-fitted scores: inside each unit the outcome labels are shuffled among that unit’s runs, holding successes and failures fixed, and A_{within} recomputed (2000–3000 permutations, two-sided). Conditioning on the observed unit structure handles unequal unit sizes with no distributional assumption; we do not claim exactness for the full learned pipeline, whose scores depend on other folds’ labels. A non-rejection is not proof of absence: for the C4 probe the effective resolution is near $A_{\text{within}} \approx 0.55$ (§5.3), so we call early probe discrimination *weak* rather than absent.

Power, comparisons, corrections. Beside each p -value we report the permutation null’s SD, which also has a closed form, $\sqrt{\sum_u P_u N_u (P_u + N_u + 1) / 12} / \sum_u P_u N_u$, reproducing the empirical values to within 0.001 on every corpus—so power for any repeated-run benchmark is predictable from its outcome table too (supplement). Predictor comparisons bootstrap the *difference* on shared resamples; effect sizes are Cliff’s delta; families of within-task tests are Holm–Bonferroni corrected at $\alpha = 0.05$.

3 BACKGROUND AND RELATED WORK

Two literatures that do not meet. Two camps report the same metric while answering different questions.

Task-level prediction estimates the probability of success from static properties of the task before any execution, for benchmark design, curriculum construction and triage [15, 20]. It is openly a model of task difficulty: Ge et al. [15] explicitly take the “majority ($\geq 50\%$) outcome” per agent–task pair, deliberately collapsing run-to-run variation.

Trajectory-level prediction estimates the same probability from the unfolding run, for monitoring and intervention [9, 23, 31, 48]. A related strand characterises the behavioural correlates of success without building a predictor from them [5, 26, 51], establishing trajectory-level analysis as an SE problem in its own right.

On a direct reading, Ge et al. [15] cite no trajectory-based failure prediction, and the trajectory papers we surveyed evaluate only against each other: we found no work reporting the control condition “ignore the run, estimate task difficulty.”

A larger body of work localises failures *after* a run completes [10, 42, 49, 50], with low accuracy (best: 53.5% agent-level, 14.2% step-level on Who&When); it is complementary, since post-hoc attribution cannot intervene in a run that has ended.

3.1 Probing the agent’s internal state

Closest to our hidden-state analysis is Silva et al. [35], whose corpus we use as C4: they show coding-agent representations encode current and future program properties (up to 0.83 AUC). Our question is complementary—whether activations separate success from failure *among repeated attempts at the same task* at deployable prefixes—and the two coexist: our controls decode length and repository identity from the very activations whose outcome signal is at chance.

The strongest published claim against a behavioural null is Ruan et al. [31]: per-round hidden-state probes said to anticipate failure from the first interaction round, with an abort cascade built on top, evaluated pooled on comparatively weak open models. We do not read our results as contradicting

it. §5.3 shows on C4 that the same probe class attains within-task 0.509 and no advantage over a three-feature length baseline—activations encode task difficulty (length recoverable at R^2 up to 0.98), enough for a pooled score, not for the abort decision. And §5.5 finds early within-task signal is detectable on C1’s weaker models and not on C2’s strong one, so a positive early result on weak open models and our null on strong ones sit together under §5.6’s trend. What the decomposition adds either way is the question a pooled evaluation cannot answer: how much of the score is the task. Direct replication needs their unreleased rollouts.

Cost-aware early stopping. A parallel line treats aborting as a decision problem, halting a run when continuation is judged unprofitable [11, 31, 34]. The savings these systems report are real; our contribution is a control condition. A policy conditioning only on the task’s historical failure rate—reading no trajectory—dominates the trajectory-based policies we test at every studied budget in replay (§6), and the classical index-policy result for costly search [44] does not transfer, since our arms retire on success and probabilities are estimated.

Concurrent work, EarlyEval [33], halts a benchmark run once a calibrated success/failure classifier is confident and substitutes the predicted outcome, preserving aggregate resolve rates and rankings across three benchmarks. The distinction matters for reading our critique: a benchmark-efficiency system (*Goal B*) only needs the *aggregate* outcome distribution, so it can succeed even when its predictor’s power is largely task-level—exactly the regime we document—whereas our analysis targets claims that trajectory evidence supports intervening on the *current* attempt (*Goal A*: abort, restart, reallocate), which requires run-specific discrimination. A large task-level component does not invalidate the benchmark-efficiency objective. §5.5 tests its released pipeline directly: the power is task-level.

Concurrent systems now report Goal-A gains. FailFast-RestartSmart [40] restarts on a 0.6B prefix monitor’s alarm, but its own control locates the gain in the intervention design, not the trigger: a *cold* restart leaves Verified resolution essentially unchanged (66.6 $\rightarrow$ 66.8%) while restarting with the interrupted diff as an overlay reaches 71.8%; the trigger is evaluated pooled, and its *run-level* contribution, against §6.3’s requirement curve, is what a within-task evaluation would settle. Telemetry monitors with rollback repair [11] beat a resampling control on small open models; rule-based enforcement [41] predicts nothing and is orthogonal.

3.2 Prior results on early prediction

Positive results exist in the uncertainty family too: TRACER [38] aggregates trajectory risk on τ^2 -bench [4], improving failure-prediction AUROC up to 37%—evaluated pooled, one run per task, so this paper’s estimand question applies directly. Two recent studies report weak early prediction signals in their respective settings. Li et al. [23] find that no uncertainty signal exceeds AUROC 0.60 at 50% trajectory progress on deep-research tasks, attributing this to *path switching*. Lee [22] exploits repeated runs on a tool-use benchmark and find the deviation signal statistically indistinguishable from zero at 10%, 25% and 50% of trajectory. The two studies address different domains and reach a consistent conclusion.

Our work is closest to Lee [22], whose mixed-outcome identification strategy we adopt unchanged and claim no novelty for. The questions differ: both prior studies ask what *causes* agents to fail and answer with mechanisms (canonical-path deviation; path switching); we ask what failure *predictors* measure—a mechanism explains why signal is absent early, not why reported accuracies are nonetheless high, which is a property of the evaluation rather than of the agent. Two scoping notes: Lee reports a *positive* within-unit effect at trajectory completion, which we do not dispute (our claims concern prefixes, where intervention is still possible); and neither study evaluates a probe of the acting model’s internal state, the strongest objection to a behavioural null, which §5.3

Table 1. The three corpora (permissive licences). Mixed-outcome = share of all units; common- r standardized rates in the supplement.

	C1	C2	C3
Source	SWE-agent [46]	SWE-rebench [3]	LiveClawBench
Scaffold	SWE-agent	OpenHands	OpenClaw
Domain	code	code	10 domains
Trajectories	80,036	67,074	6,834
Tasks	3,591	6,306	134
Models	3 (Llama-3 8B–405B)	1 (Qwen3-Coder-480B)	17
Failure rate	0.833	0.521	0.582
Mixed-outcome	22.0%	27.6%	21.9%
Same-unit weight w	2.5×10^{-4}	3.4×10^{-5}	8.8×10^{-5}
Mixed units	926	1,741	500
Within-unit pairs	220,055	38,576	1,000

addresses. The recommendations differ too: Li et al. [23] advise final-step confidence for restart decisions over in-trajectory intervention; our results likewise do not support the studied signals justifying in-trajectory intervention in these settings, while §6 shows a policy conditioning on task difficulty alone—no trajectory signal at any point—outperforms every trajectory-based policy we tested in replay.

3.3 A gap in the literature

Four Google Scholar query families (2023 onward), every returned trajectory-level failure-prediction result read and snowballed [7, 9–11, 15, 21–24, 31, 33, 38–40, 42, 49–51]; search protocol and verification discrepancies in the supplement. The retrieved papers divide on five properties, each checkable from a method section and each load-bearing for whether a predictor could be deployed as a monitor: operating on a partial trajectory; repeated runs of the same model on the same task; naturally occurring failures; breadth of evaluation environments; and measuring the effect of acting on the prediction (the full property map is a supplement table). We read that map descriptively rather than as a uniqueness claim. What it shows is that the pooled-versus-within estimand question has not been made load-bearing anywhere in this literature, and that among the prefix-level works with repeated runs, none measures the effect of acting on the prediction.

One entry deserves comment. Ruan et al. [31] collect 8 rollouts per task (100 TextCraft tasks; 200 WebShop at 4) and split grouped by task—precisely the design the within-task estimand requires. What is reported is pooled AUROC; the within-task quantity is never computed. The pattern this paper is about is visible in the work best placed to avoid it.

4 STUDY DESIGN

4.1 Corpora

We need benchmarks recording several independent attempts at the same task by the same model. Three public corpora with many runs per unit qualify (Table 1); a fourth, built for the probe requirement, is §4.2. C3’s 6,834 runs parse to 6,727 (2,269 units; 488 parseable mixed—the permutation-test set). The corpora differ in scaffold, benchmark, model family, domain and base failure rate, which is what lets the cross-corpus question be answered.

The observed mixed-outcome rate—the fraction of (model, task) units producing both a success and a failure under the available repeat counts—is between 21.9% and 27.6% in all three corpora

despite their differences. For latent success probability q and r attempts the observed rate is $1 - q^r - (1 - q)^r$, which grows with r , so these are not intrinsic “flakiness rates”; but at matched small r the consistency across corpora says a substantial fraction of units is not determined by task and model alone.

C3 additionally carries *independently annotated* difficulty tiers and domain labels—author-produced, not outcome-derived, hence a stronger instrument than any oracle; their validity shows in failure rising monotonically $0.313 \rightarrow 0.710 \rightarrow 0.962$ across tiers.

4.2 A fourth corpus, for the activation probe

The hidden-state probe requires the *acting* model’s weights, and most public corpora record proprietary-model trajectories. We use the LATENT PROGRAMMING HORIZONS corpus of Silva et al. [35] (C4): SWE-bench trajectories from two open mixture-of-experts models, LAGUNA-XS.2 (40 layers, $d_{\text{model}} = 2048$, $\approx 33\text{B}$ total/ $\approx 3\text{B}$ active) and QWEN3.6-35B-A3B. Each record stores the *exact token identifiers* consumed, with per-message offsets, so teacher-forced replay is exact by construction—no chat-template reconstruction, a common and undetectable error source in probe reproductions.

Both splits record ten attempts per task on 500 SWE-bench tasks; mixed-outcome rates (40.6%/34.8%; 203/174 mixed units) exceed C1–C3; C4-L’s median trajectory is 35,759 tokens. Three C4-L unit counts recur in this paper and differ by design, not inconsistency: 203 mixed units in the full behavioural corpus, 192 when C4 is featured identically to C1–C3, and 142 in the probe subset, whose 16,000-token cap excludes the longest units (sensitivity in §7).

4.3 Probe construction

We cut after assistant turn $k \in \{1, 3, 5, 10\}$, take the last emitted token’s hidden state at layers $\{10, 20, 30, 40\}$, and probe with ℓ_2 -regularised logistic regression on the raw 2048-dim activation—linear deliberately, the probing literature’s own standard for explicit representation. Every fit uses GROUPKFOLD by task (no task in both train and test), with regularisation selected by nested GROUPKFOLD inside each training fold, never on test data. Prefixes reach 40,133 tokens at $k=10$; we do not truncate below the maximum, since a length cap induces outcome-dependent selection (§4.5).

4.4 Predictors and the difficulty oracle

Trajectory predictors read the first k turns: behavioural features (turn counts, message/observation lengths and dispersion, error tokens, repetition, tool diversity) plus TF-IDF text, fit by logistic regression and gradient boosting, out-of-fold under GROUPKFOLD on task. Label-only predictors (C3) use pre-execution categorical facts—difficulty tier, domain, model—and never observe the run. The difficulty oracle scores each run by its unit’s failure rate, leave-one-out for pooled figures.

One caution: a leave-one-out group mean is spurious for the *within*-task figure, because inside a unit its only variation is minus the run’s own outcome, which gives $A_{\text{within}} = 0$; we therefore use the plain group mean, constant within unit, yielding 0.5. Several distinct “task difficulty” notions with different deployment assumptions recur; the supplement fixes the terminology.

4.5 Prefix definition, and a confound

A monitor observes a run turn by turn without knowing how long it will last. Our own initial design featured a fixed *fraction* of each trajectory and included k/T , where T is the final turn count—unknown at monitoring time and outcome-correlated. On C1 that single feature inflates A_{within} by 0.044–0.074 depending on depth, and removing the remaining length features on top

Table 2. Decomposition on C1 (SWE-agent). Absolute-turn design, confound removed. The oracle row uses only each unit’s failure rate—no run-level information—and covers all 80,036 records; featurised rows cover the 79,929 parseable trajectories.

Condition	n	A_{pool}	A_{within}	gap	units
$k = 2$	79,929	0.6435	0.5169	+0.127	926
$k = 5$	77,951	0.6863	0.5428	+0.144	920
$k = 10$	63,657	0.7219	0.5987	+0.123	885
$k = 20$	35,554	0.7463	0.6527	+0.094	620
Difficulty oracle	80,036	0.9454	0.5000	+0.445	926

changes little, so it acts alone. All results in this paper use absolute turn indices with the feature removed; §5.4 measures the same leak in a published protocol.

5 RESULTS

Roadmap: §5.1 answers RQ1; §5.2–5.6 answer RQ2 (measurement; objections; a published predictor; mechanism); RQ3 is §6.3.

5.1 RQ1: what pooled AUROC measures

On C1 (Table 2), early run-level discrimination is near chance (0.517 at two turns, 0.543 at five). It becomes modest only late, and only on the 44% of runs that survive twenty turns; the gap is +0.09 to +0.14 throughout.

The oracle row makes the ceiling concrete: knowing only each unit’s failure rate, and nothing about the run, reaches pooled 0.945 while at chance within task. This is not mere oracle leakage. A *deployable* pre-execution predictor—issue-text TF-IDF plus repository identity, tasks held out, no target-task outcomes—reaches pooled 0.598 (C1) and 0.616 (C2) while still scoring 0.500 within task.

Answer to RQ1. In benchmarks that record repeated attempts, pooled AUROC is dominated by cross-task pairs ($w \leq 6.8 \times 10^{-4}$ here) and so measures task-difficulty estimation: a scorer that knows only each unit’s failure rate reaches pooled 0.945 on C1 at exactly 0.5 within task, and a deployable pre-execution predictor reaches 0.598–0.616 at the same within-task chance level. The pooled figure cannot distinguish a predictor that reads the run from one that estimates the task.

5.2 RQ2: run-level information, measured directly

C3 answers this most cleanly: its labels are independent annotations, so we lead with them and treat the leave-one-run-out oracle strictly as a ceiling (held-out pre-execution prediction reaches 0.841 on SWE-bench Verified [15], matching our task-only 0.826).

The result is stark, and a property of the *task*, not the model. Task metadata alone—difficulty tier and domain—reaches pooled AUROC 0.826 (within-task 0.5); adding model identity lifts it only to 0.845, and model identity alone reaches just 0.568. Removing the model confound entirely, within *each* of the 17 models separately the per-model task oracle reaches pooled median 0.985 (range 0.962–0.998), still above the 0.85–0.94 band with the model held fixed. Trajectory predictors top out at 0.732–0.752 pooled, within-task 0.498–0.542 (covering 0.50 at two of three depths). Pooled AUROC would select a predictor with no demonstrated ability to do a monitor’s job.

The variance ceiling. On C3, a finite-repeat-corrected variance-components analysis attributes $\approx 70\%$ of outcome variance to between-unit heterogeneity (an ANOVA ICC(1) and a beta-binomial method-of-moments correction agree at 0.699, 95% bootstrap CI [0.68, 0.72]; the naive $\text{Var}(\bar{y}_u)/\text{Var}(y) = 0.80$ is inflated by within-unit sampling noise at ≈ 3 runs per unit). This is not an AUROC bound, but it says most of the variation a pooled evaluation sees is between-unit variation, which is why a label-only predictor does so well on it. The direct measurement of what a predictor gains follows.

What does the trajectory add? The direct measurement: three nested cross-fitted predictors on C3. M_0 knows only the task (its unit’s failure rate from that unit’s *other* runs); M_1 only the trajectory; M_2 both. Pooled AUROC/Brier/log-loss: M_0 0.894/0.103/0.350;¹ M_1 0.726/0.211/0.606; M_2 0.923/0.103/0.340 (6,727 runs, 2,269 units).

The task oracle thus lands inside the published 0.85–0.94 band. Note the conditioning: M_0 is not a generalising task classifier but a repeated-run estimate, so the increment reads “once historical outcomes from the same task–model unit are available, adding the entire current trajectory improves pooled AUROC by +0.029” [+0.023, +0.034]; it improves log loss slightly (+0.0101), and leaves Brier unchanged (+0.0007, [−0.002, +0.003])—most visible gain is in ranking rather than probability refinement. Within task, M_0 is 0.5 by construction; the trajectory features alone reach 0.587 on this corpus (which pools all seventeen models, including the weak ones whose failures §5.6 shows are legible); a within-task value for M_2 is not identifiable under leave-one-out task scoring (the caution of §4.1).

Replication across scaffolds and domains. C2 (SWE-rebench / OpenHands v0.54.0; 67,074 trajectories over 1,823 repositories, failure rate 0.521) produces the cleanest null in the study. A_{within} is flat at 0.4935–0.5014 from $k=2$ to $k=20$ with every interval covering 0.50, the median unit’s own AUROC is exactly 0.500, and only 46–48% of units lie above chance, a pattern strongly consistent with a tightly concentrated null at our resolution. Meanwhile the gap holds at +0.16 to +0.18 (pooled 0.660–0.672; difficulty oracle 0.974).

A difference between corpora. C1’s within-task accuracy rises with depth (0.517 $\rightarrow$ 0.653) where C2’s does not (0.494 $\rightarrow$ 0.497); re-running C1 with C2’s exact eight features changes its curve by at most 0.035, so featurisation does not explain the divergence—§5.6 offers the reconciliation.

Hypothesis tests. Permutation tests of $H_0: A_{\text{within}} = 0.5$ (§2.5) reject in no condition after Holm: C2’s null is tight (± 0.0057 ; a true 0.511 would flag) with observed 0.4934, and C3’s one nominal hit ($k=5$, $p = 0.027$) misses the threshold. On C3’s own pooled metric the label-only predictor beats the trajectory predictor by +0.134 [+0.120, +0.148]: **a predictor that never observes the agent wins on the field’s metric** while exactly at chance on the run-specific estimand (tables in supplement).

5.3 RQ2, the objections: probes, length, power, and labels

The strongest objection to a behavioural null is that the signal is latent in the acting model’s internal state [31, 35]. We test it on C4 by teacher-forced replay from the exact stored token ids, probing layers {10, 20, 30, 40} at prefixes $k \in \{1, 3, 5, 10\}$ with cross-fitted logistic probes (16 configurations per model). Across the 16 cells on C4-Q (174 mixed units) the mean within-task AUROC is 0.509 (median 0.513, range 0.458–0.551); one cell reaches 0.551 (Holm-adjusted $p = 0.048$; 95% CI [0.503, 0.588], at the resolution boundary of §2.5). On C4-L no cell survives (best $p = 0.40$, mean 0.513). The probes never beat a three-feature length baseline (+0.043, 95% CI [−0.014, +0.099]),

¹ M_0 (cross-fitted) reads 0.894; the direct outcome-table oracle of §2.3 reads 0.903 on C3—the same ceiling, differing only by finite-sample estimation.

while that same baseline attains the highest *pooled* AUROC of any predictor we test at every depth—and the activations demonstrably encode length (R^2 up to 0.98) and repository identity (AUROC up to 1.00), so what little the probes recover is plausibly length re-expressed. Probes over these representations reveal no strong early within-task signal, and the probe class is not the reason: a one-hidden-layer MLP moves the estimate by under 0.005 (0.519/0.526 at $k=1/10$) and mean-pooling states across cut points gives 0.484. Modest effects near the resolution boundary of §2.5 cannot be excluded. The full 16-cell grids with controls are in the supplement.

Can this pipeline detect an effect that is there? Running the identical pipeline on known answers: an outcome-carrying feature is recovered at within-task 1.000, 32 Gaussian-noise features give 0.503, and—the load-bearing check—*real activations with shuffled labels* give 0.488. The machinery reporting 0.524 on real labels returns 1.0 when signal exists and 0.5 when not; the estimator matches an independent implementation to 10^{-9} , and the permutation null centres at 0.502 ± 0.021 .

Trajectory length as the dominant signal. On pooled evaluation the length baseline does not merely compete—it wins. Three scalars (absolute token position, its log, assistant-turn count) beat the best 2048-dimensional activation probe at every depth on C4-L: pooled 0.710/0.712/0.707/0.706 against 0.639/0.619/0.652/0.677 at $k = 1, 3, 5, 10$ (142 units, full table in the supplement). Its *within*-task value is 0.502–0.503: length carries strong between-unit information while providing essentially no within-unit discrimination—this paper’s point in miniature, and independently observed by Mehtiyev and Assunção [26], whose length signal reverses under a within-task control.

Bounding the null: equivalence and power. On the full C4-L sample (142 mixed units) the bootstrap SE of A_{within} is 0.0193; the 90% interval [0.491, 0.556] establishes equivalence to chance at $\delta = 0.10$ but misses $\delta = 0.05$ by 0.006, and the minimum detectable effect at 80% power is 0.054: we can exclude $A_{\text{within}} \geq 0.554$ and cannot exclude 0.53. The inflation gap itself is significant and tightens with sample: $A_{\text{pool}} - A_{\text{within}} = +0.107$, 95% CI [+0.046, +0.165]. The weighting of A_{within} does no hidden work: on C4-L unit-uniform macro-averaging gives 0.5243, task-demeaned 0.5173, min-repeat thresholds [0.5229, 0.5241]; and the pattern holds on the other corpora at $k=10$ —C1 pair/macro/deployment-weighted 0.587/0.581/0.576, C2 0.499/0.495/0.495—every variant on the same side of chance as the reported one.

Is the null a label-noise artifact? Our labels are the benchmarks’ own oracles, and Wang et al. [43] report that 7.8% of “solved” SWE-bench patches fail the developer suite (6.2pp resolution inflation). Noise attenuates AUROC toward 0.5. Because a deployed pipeline is *trained* on the corrupted labels, attenuation compounds beyond the fixed-scorer formula (0.15 AUROC at $\alpha = 0.08$ where that formula predicts 0.03), so we calibrate it empirically: a true 0.55 falls to our observed 0.524 at $\alpha = 0.08$, almost exactly the documented mislabelling rate; a true 0.60 requires $\alpha = 0.49$; 0.65 never reaches it. Effects of 0.60 and above are excluded; effects near 0.55 are not, converging with the power analysis by an independent route. One property survives: attenuation is multiplicative in $A - \frac{1}{2}$, so noise can shrink a real effect but cannot manufacture our inflation gap (sweep tables in supplement).

A verification protocol for probe studies. Four gates, each of which caught a silent pipeline failure: (1) weight integrity (a teacher-forced accuracy floor caught one code path silently randomising all MoE layers); (2) replay from stored token ids; (3) an independent estimator agreeing to 10^{-9} ; (4) the positive-control battery above (full protocol in the supplement).

5.4 RQ2 on a published predictor

Perhaps published methods reaching 0.94 have run-level signal our predictors miss. We reimplement Automata [9] from its description (artifacts unreleased; most of their corpora lack repeated runs): an activity-abstracted automaton with gradient boosting over per-state behavioural and anomaly features, transition model fit on training folds only; the full recipe and FSM-size checks are in the supplement.

The reimplementation reproduces their reported result. Cho et al. [9] are explicit that their headline 0.941 is tau2-bench telecom; for SWE-agent, C1’s scaffold, they report 0.799. Our reconstruction reaches 0.806 pooled on full C1 trajectories—within 0.007, same scaffold, FSM size inside their reported range—so the 0.941 a reader might expect us to match was never a SWE-bench number. The ceiling formula sharpens both directions from public outcome tables alone, and the answer is cohort-dependent: τ^2 -telecom [4, 47] admits difficulty-only ceilings of 0.67–0.87 for strong leaderboard models failing 9–13% of the time, but 0.93–0.98 for the four baselines the benchmark itself releases, which fail 27–66% of the time (supplement): whether a 0.941 there is task difficulty depends on the acting model’s regime, §5.6’s conditionality inside one benchmark. C1’s table admits 0.954, above both SWE-agent figures (checked models need not match theirs). Because A_{pool} and A_{within} come from the same cross-fitted scores, their relationship is internally valid for our reconstruction; §5.5 closes the remaining gap by scoring a published system’s own released code.

Early, essentially no run-level signal (figure in the supplement): at one turn $A_{\text{within}} = 0.507$ vs $A_{\text{pool}} = 0.565$ —within resolution of chance exactly where early abort would be worth most. **Late, it is real:** 0.603 by ten turns, 0.688 at full trajectory; what the pooled figure obscures is *when* prediction becomes possible. **The gap persists at every depth** (+0.057 to +0.126), growing as the predictor improves.

The fractional checkpoint leaks, and we can measure how much. The triangle in the supplement’s figure is that paper’s own online protocol at 25% of the trajectory: a mean of 6.7 observed turns, yet a higher score than fixed-turn protocols observing as much or more. The source is *look-ahead leakage*: a fractional checkpoint depends on eventual length T , unavailable online and outcome-associated.

We measure the leak three ways (supplement): T alone reaches within-task 0.6138; a fixed-turn control observing marginally *more* (6.8 vs 6.6 turns) scores 0.5658 against 0.6297—the +0.064 is the leak; and decoupling content from outcome within every unit still leaves 0.6180, so the surplus is leaked length, not content.

None of this criticises the method; it is a property of the protocol, and applies to any system evaluated at fractional checkpoints—including both prior negative results [22, 23], whose early figures are by this measurement if anything optimistic. The remedy is cheap: cut at a fixed turn count or token budget—on C1, $A_{\text{within}} = 0.5658$ and 0.5710, close to each other and far below the fractional 0.6297: what a leak-free protocol looks like.

5.5 RQ2 on a published system’s own released code

Reimplementation leaves a residual objection: perhaps our reconstruction, not the method, lacks run-level signal. EarlyEval [33] closes it, because its pipeline is released. We run that pipeline *unmodified* on C2 with an instance-holdout split and its default gold-answer features (task-constant, hence able to raise only pooled discrimination). The evaluation was pre-registered as a five-fold instance partition, above the system’s own published scale, and every figure below aggregates the five folds by summed pair counts as registered, so each of C2’s 6,306 instances is tested exactly once (223 mixed units, 4,935 within-unit pairs; supplement).

Its checkpoints index *steps observed*, so index 0 predicts before the agent acts. At that index the system already reaches pooled AUROC up to 0.72 from the task prompt and metadata alone, while within-task is exactly 0.500 in 70 of 70 model-fold combinations—the value a task-constant scorer must take, and a check that the estimand behaves as derived. In other words, publishable-band pooled discrimination needs no run information at all. Once run content enters, it is not used. At one observed step the feature vector demonstrably differs across attempts (action text in 96.7% of instances, thought text in 99.5%), yet within-task AUROC is 0.49–0.52 over all 14 variants and stays at chance through five steps (0.45–0.52; the only early aggregate tests reaching $p < 0.05$ sit below chance, at 0.456–0.475). By run end, pooled reaches 0.77 while within-task never exceeds 0.58. The pre-run interval is degenerate at $[0.500, 0.500]$ (permutation null SD exactly 0) and the minimum detectable effect is 0.55, so every early interval excludes any value the requirement of §6.3 would care about. With $w = 0.001$, 99.9% of the system’s pooled pair mass is cross-task: it serves Goal B, and under the Goal-A estimand its discrimination is task-level.

A second released system, opposite paradigm. The released one-class monitor of Dubey [11] is fitted on *healthy runs only*, so no failure-label objection reaches it. Run unmodified on the same fold and test instances it transfers weakly (pooled 0.44–0.55 through 30 steps), so its early null is uninformative; its modest late signal, though, decomposes the opposite way to EarlyEval’s—pooled 0.58–0.60 against within-task 0.55–0.57 (supplement).

A weaker-model corpus, where early signal does appear. C2 carries one acting model, so we ran the same released pipeline on C1 (SWE-agent scaffold; Llama-3 8B/70B/405B), whose assistant turns carry a fenced command we map to its action schema. Pre-run behaves as on C2—within-task 0.500, pooled up to 0.63. Early behaviour does *not*: from one observed step within-task runs 0.49–0.56 (3 of 6 permutation tests below 0.05), reaching 0.50–0.57 by five steps (5 of 6) and 0.50–0.64 at run end. On C1, unlike C2, early run-level signal is detectable—the direction §5.6’s trend predicts, since C1’s models are the weaker ones. We therefore report an early null for *strong* acting models and a weak but real early signal for weak ones, still far below §6.3’s requirement. Two caveats: 48 of the 54 mixed test units are one model (Llama-70B), so this is largely a single-model result despite the corpus spanning three; and late checkpoints thin the cohort (26 units by 30 steps), so we do not read them.

5.6 Mechanism: depth, capability, and the probe null

Signal accumulates with depth, and it is not selection. On a *fixed cohort*—the 63,657 C1 traces with ≥ 10 turns, identical at every depth—the automaton featuriser gives 0.509/0.511/0.532/0.586 at $k = 1, 3, 5, 10$ (885 units), so early discrimination sits within resolution of chance on a constant trace set and is clearly above it by ten turns. Survivor selection can only inflate a curve that rises, which among our corpora is C1 alone. Nor is the pooled training objective the reason: a pre-registered same-unit conditional-logit ranker lifts within-task to 0.526/0.535/0.547 at $k=1/3/5$ against 0.504/0.509/0.537 pooled ($p < 0.002$)—detectable, small, far below §6.3’s requirement—while its pooled AUROC collapses toward chance: stripped of difficulty, little is left (supplement).

Legibility falls as the acting model strengthens. Across C3’s 17 models (one predictor, cross-fitted by task): $\rho = -0.662$ ($p = 0.0038$); the weakest model’s failures are identified at 0.832, the two strongest at 0.444 and 0.490. Within lineage, 15 of 19 same-family pairs show the stronger member less legible ($\rho = -0.749$). The association survives permutation, matched pair counts, disjoint capability measures and leave-one-family-out (supplement), and the direction replicates on C1’s second scaffold (Llama-8B 0.636 > 70B 0.581 > 405B 0.569 at ten turns) and again in §5.5, where a *released system*’s late within-task runs higher on C1’s weaker models than on C2’s strong one.

It remains an association, not a causal claim: runtime prediction *may* grow harder as models strengthen. This is task-outcome legibility from behavioural summaries—a narrower channel than safety monitorability [17]; no contradiction, a different channel.

The obvious explanation does not hold: behavioural dispersion [2] falls with capability ($\rho = -0.591$), echoing Gould’s shrinking-variance account [16], but partialling it out changes nothing ($-0.662 \rightarrow -0.668$). Instead the signal vanishes from every behavioural channel at once—each family separately identifies failure in the weakest tercile (0.61–0.64), none clears chance in the stronger two—consistent with weak agents exposing failure through coarse procedural cues that stronger agents’ failures do not leave; the underlying mechanism remains open (per-channel tables in the supplement).

This reconciles the corpora and the probe. C2’s single, much stronger model (Qwen3-Coder-480B-A35B) stays at chance at every depth (0.500–0.504; 1,741 mixed units, 38,576 pairs, a well-powered null with no selection, every trace exceeding ten turns) where C1’s weaker Llamas climb to 0.586. And the C4 probe null is the *expected* observation: featurising C4’s recorded trajectories exactly as C1–C3 (matched-depth table in the supplement), behaviour is also at chance at the probed depths (0.501/0.496/0.514; 192 mixed units, no token cap) and reaches 0.656 only at full trajectory. Nothing is legible to either channel that early. A monitoring policy should not assume predictability transfers from another deployment; it must be established for one’s own model and scaffold.

Answer to RQ2. Run-specific information at online-computable prefixes is weak. On strong acting models, early within-task AUROC sits at or near chance for behavioural predictors, a reimplemented published monitor (0.507 at one turn), a released pipeline (0.49–0.52 at one step) and hidden-state probes (mean 0.509), all within our resolution boundary of ≈ 0.55 . Signal accumulates with depth (0.586 at ten turns on a fixed C1 cohort) and is larger for weaker acting models ($\rho = -0.66$ across 17 models; 0.57 by five steps on C1), but no early value approaches what intervention requires (RQ3).

6 RQ3: DEPLOYMENT CONSEQUENCES

A critique that stops at “the metric is wrong” leaves practitioners nothing. This section therefore gives a diagnostic to report and a policy that performs strongly in replay—at its simplest, working through tasks in a difficulty order borrowed from another model—in the repeated-evaluation settings SE teams already run: benchmark sweeps, CI re-execution and model migration.

Report the gap. Report ($A_{\text{pool}}, A_{\text{within}}, w$) and $\Delta = A_{\text{pool}} - A_{\text{within}}$: $\Delta \approx 0$ means the predictor reads runs; $\Delta \approx A_{\text{pool}} - 0.5$ means unit identity and nothing else. Here Δ ranges +0.09–+0.47, at no cost beyond runs a repeated-attempt benchmark already has.

6.1 A Bernoulli-arm reduction

Within a unit: no run-level signal above resolution (§5.3); cost independent of outcome given task ($p = 0.18, 0.63$; paired difference $< 0.7\%$ of mean cost); ten temporal diagnostics (lags 1–3 against a permutation null carrying the $-1/(n-1)$ bias, runs tests, half-splits) show one marginal signal ($p = 0.045$ uncorrected). None of this *proves* exchangeability; we treat exchangeable Bernoulli(θ_u) outcomes as an approximation the diagnostics do not contradict, and evaluate by replaying real runs, not the model. Under it, allocation is a budgeted multi-armed bandit: each task an arm, each pull costing random c_u , a token budget B limiting pulls, an arm retiring on success. Retirement is endogenous—outside both the budgeted bandit [45] and mortal-bandit [8] analyses—so no regret guarantee is claimed. In the fluid limit, expected cost to solve u is $\bar{c}_u/\theta_u$, so greedy $\theta_u/\bar{c}_u$ maximises solves under budget—a ratio-index policy (cf. Smith’s rule [36]) the oracle implements and Thompson sampling approximates (supplement).

Budgeted Thompson sampling with a transferred prior. Budgeted Thompson Sampling [45], Beta-Bernoulli: sample $\tilde{\theta}_u$ per arm, pull $\arg \max_u \tilde{\theta}_u / \bar{c}_u$, update. The prior transfers another model’s outcomes on the same task ($\alpha_u = 1 + \lambda s_u^{\text{other}}$, $\beta_u = 1 + \lambda(n - s_u)^{\text{other}}$, $\lambda=0.5$). Replay evaluation: each pull draws a real recorded run, charges its real cost, returns its real outcome, retires solved tasks; 200 budget realisations. The contribution is reduction, prior and evaluation, not the algorithm.

6.2 Allocation results

The effect is large exactly where theory predicts: under budget pressure. On C4-Q the policy solves 162.5 tasks at 2M against round-robin’s 53.1 (3.1 $\times$, and 94% of an oracle knowing every true success rate), the transferred prior contributing +14–20%. Table 3 reports C4-L in the reverse transfer direction on a held-out split with tuned abort baselines (55.2 vs 16.0 at 1M for the ranking). The margin narrows with budget and vanishes by 40M—a sanity check, since gains in the unconstrained regime would be suspect.

A second corpus, and the cost-side reading. C4 alone records per-run cost, which is why the replay lived there; on C2 we estimate it from trajectory length (only relative cost matters here) and transfer the prior from C1—across model family, scaffold and corpus. Allocation generalises: 37.5 tasks at 2M against round-robin’s 16.8, 2.2 $\times$. Read along the other axis, the same simulation gives the statement practitioners act on: the same number of solved tasks costs 22–36% **fewer tokens**.

Where our own prior fails. Registered in advance, reported against our interest: on C2 an *uninformative* prior beats the transferred one at every budget (52.5 vs 46.1 at 2M)—C1’s models fail 83% of the time against C2’s 52%, and an ordering learned from a far weaker model mis-ranks for a stronger one. The prescription is conditional: always allocate rather than abort, transfer a prior only from a model of comparable capability, and do not route on monitor evidence at all—a second, cost-honest replay has it solving 28.3 tasks at 2M against round-robin’s 30.9 (supplement).

The ranking does most of the work. The adaptive machinery is not where the value sits: a *static* transferred ranking, walked in fixed order, *beats* the bandit at scarce budgets on both corpora (166.6 vs 162.1 at 2M C4-Q; 55.2 vs 51.9 at 1M C4-L), overtaken only at 10M (295.8 vs 307.8; 151.7 vs 161.4) where outcomes correct the prior. We recommend the ranking as default, adaptation as a refinement for loose budgets. A greedy retry-the-top variant collapses (208.0 flat)—a property of the retry rule; give-up and cost-sampling variants move counts by under one task (supplement).

Where the prior comes from. Difficulty transfers across models ($r = 0.863$; a transferred prior keeps 92% of oracle ranking power), so a prior transferred from one acting model remains effective on the other, in both transfer directions. For genuinely novel tasks, pre-execution prediction is real but not yet actionable: Ge et al. [15]’s released features predict our observed difficulty at $\rho \approx 0.53$ held out (rollout-derived references: 0.83–0.85), yet seeding the prior from them *loses* to an uninformative prior at every weight (−0.3% to −11.1%) while the rollout-derived prior gains +14.4%. We do not infer a universal correlation threshold from this; here, pre-execution estimation serves triage (Q1), not allocation (supplement).

6.3 Abort or reallocate? A direct comparison

No baseline is a strawman and every increment of information is visible: tasks split 50/50 into tuning and evaluation; abort thresholds tuned *per budget* on tuning-set replay (separately per policy); the held-out set touched once, all policies on paired seeds; allocation’s prior transferred from C4-Q, never from the replayed tasks’ own outcomes. Replay is as in §6.1, uniform *with* replacement (the posterior estimates a success probability, not a finite population), cost and outcome from the same

Table 3. Distinct tasks solved at a fixed token budget on the C4-L held-out task split (250 tasks), 150 paired replay trials, mean and 95% percentile interval. Abort thresholds are tuned per budget on the disjoint tuning split. TS is budgeted Thompson sampling over tasks; the prior is transferred from C4-Q.

Policy	1M	2.5M	5M	10M
Round-robin	16.0 [10,24]	38.6 [28,48]	76.8 [65,90]	146.9 [139,156]
Static transferred rank	55.2 [52,58]	99.5 [94,104]	134.2 [127,140]	151.7 [141,159]
TS, uninformative	47.3 [44,52]	80.9 [76,86]	115.3 [110,121]	143.7 [138,150]
TS + transferred	52.5 [48,56]	94.4 [88,100]	130.8 [124,137]	161.2 [155,167]
Tuned abort only	16.8 [10,24]	45.3 [36,55]	82.9 [72,93]	144.2 [135,154]
TS+transf. + tuned monitor	48.3 [44,52]	93.8 [88,99]	128.6 [123,136]	159.2 [153,165]

recorded run; aborting at turn k charges only tokens through k (generous: forgone late successes are free); the monitor is the cross-fitted automaton predictor at $k=5$ —exactly the measured quality.

The comparison the paper’s question turns on is the last row against TS+transferred: *given the same task-level policy and prior information, does the runtime monitor add anything?* The paired difference is between -4.2 and -0.6 tasks across the four budgets, every 95% CI spanning zero (widest $[-11.0, +7.0]$): a threshold-tuned monitor of the measured quality does not improve a strong allocation policy at any studied budget. The tuned abort-only policy is worse still— 17.2 to -49.0 tasks against allocation, all paired 95% CIs negative—and tuning is not the issue: it beats the median abort it replaces (144.2 vs 120.7 at 10M) yet never beats round-robin’s interval.

A pre-registered semi-synthetic sweep locates the *requirement*. Replaying monitors of controlled within-task AUROC at $k=5$ (tuned thresholds as above), the hybrid first beats allocation at within ≈ 0.84 (10M) and ≈ 0.93 (5M) on C4-L; even perfection is borderline at 1M ($+2.6$ [$+0.0, +6.3$]) and adds at most $+7.8$ tasks. That band is the monitor-favourable case twice over. With errors *correlated* with task difficulty ($\rho = 0.3/0.6$) the crossing rises to 0.90 (10M) and 0.90 – 1.00 (5M); and on C2, against the strongest allocation baseline available there, *no* tested level clears allocation at 2–10M, perfect discrimination included—only at 25M does a perfect monitor gain ($+9.2$ [$+4.0, +15.0$]). Headroom is small because allocation already avoids most failing pulls; it eases with budget and depends on k . Without task information abort-only never clears round-robin, and measured monitors sit below every crossing in all families (supplement). Turn control—the strongest SE-specific compute-control baseline [14]—does not close the gap either: capping successful-run length at the 50/75/90th percentile leaves round-robin far behind allocation (15.7 vs 37.9 at 2M on C2), and capping *on top of* allocation costs tasks at every budget (-0.1 to -53.8), because a good allocator has already avoided most failing pulls (supplement).

The operating curve is concrete: retaining 99/97/95% of C4-L successes at $k=5$ catches only 1.4/5.4/9.1% of failures and saves 1.2/4.3/7.3% of compute. In the C1 abort-and-restart simulation (supplement), metric choice changes which policy appears deployable [39].

Answer to RQ3. Yes. By pooled AUROC a difficulty-only scorer would be selected as the best monitor; on the estimand the abort decision needs, it is at chance. In replay, allocation by a static difficulty ranking transferred from another model solves up to $3.5\times$ more tasks than round-robin at scarce budgets, a tuned abort-only policy solves 11–68% fewer than either, and adding the measured monitor to allocation changes throughput by -4.2 to -0.6 tasks with every 95% CI spanning zero. A monitor would need within-task AUROC ≈ 0.84 – 0.93 before aborting adds throughput; measured early values are 0.50 – 0.55 .

7 THREATS TO VALIDITY

Construct validity. Labels are the benchmarks’ own oracles, and these are imperfect: 7.8% of “solved” SWE-bench patches fail the developer suite [43] and 10.7% of passes are flawed [1, 32]; label noise is a known SE confounder [37]. §5.3 bounds its effect: noise can shrink a real within-task effect but cannot manufacture the inflation gap, and effects of 0.60 and above are excluded.

Internal validity. Runs within a unit may not be exchangeable; ten diagnostics find one marginal signal ($p = 0.045$), and the allocation results replay real runs rather than the model. The probe’s 16k-token cap excludes the 129/500 longest C4-L units, but the estimate is stable as they enter ($0.498 \rightarrow 0.513$). Our Automata reconstruction could differ from the original; §5.5 has no such step, since it runs a released pipeline. Late C1 checkpoints thin the survivor cohort and are not read.

External validity. The corpora are coding-heavy, but C3’s eight non-coding domains still show the difficulty oracle at 0.909, and a pre-registered outcome-table analysis of τ^2 -bench’s four non-coding domains (10,832 runs, no trajectories or predictor) finds mixed rates, w and difficulty-only ceilings each comparable to our corpora, so the precondition is not peculiar to coding (supplement). The probe corpus is two open mixture-of-experts models of similar scale, and the capability association rests on one primary scaffold, replicated on a second. The allocation method needs prior attempts at the task (§6.2), rests on tested approximations (§6.1), carries no regret guarantee because retirement is endogenous [8, 45], and is evaluated in replay, which cannot capture live interactions.

Statistical conclusion validity. A_{within} is identifiable only on mixed-outcome units (21.9–40.6% of units), a principal stratum [13]. A deployed monitor cannot know it is on such a unit, so the conditioning is not available online. It is, however, the stratum where the decision has consequences: on a unit that never or always succeeds, any abort is respectively costless or pure loss, and what a monitor would need is the *task’s* rate—the difficulty signal already in the allocator (§6.3). The conditioning thus scopes the estimand to runs where run-level discrimination could change an outcome. Nor does it drive the result: four re-weightings—including a posterior-predictive one over all 950 units that conditions on nothing the sample showed—move early within-task by at most 0.027, the unconditioned weighting sitting *closer* to chance than our primary (0.506 vs 0.516; supplement). Evidence per unit is thinnest where generality is argued: C3’s $r=3$ yields 2.0 within-unit pairs per mixed unit, so its *within*-task values carry wide intervals (its non-coding oracle figure is pooled and unaffected). Multi-model corpora share tasks across units, so the capability trend uses a model-resample bootstrap. Probe resolution is ≈ 0.55 (§2.5); grids are Holm-corrected, except the released-pipeline early tests, which we report as a rate against the count expected by chance.

8 CONCLUSION

Agent failure predictors report AUROC of 0.85–0.94, read as evidence that a running agent can be monitored. That number cannot distinguish reading the run from estimating the task: the same-unit weight is $\Theta(1/n)$, the difficulty-only ceiling is computable from the outcome table, and a leave-one-run-out oracle reaches 0.894 while unable to tell two attempts apart. Measured directly, early run-level discrimination is absent on strong acting models and weak on weaker ones, and in replay task-level allocation outperforms every abort policy we tested.

For researchers, a benchmark should report the estimand its decision needs: A_{within} and w against difficulty baselines, on prefixes cut at online-computable indices (§5.3). For practitioners, allocate before aborting, and establish predictability for your own model and scaffold. Monitoring is not worthless—signal exists late and for weaker models—but the pooled metric cannot say whether it exists for yours, and the same decomposition applies wherever observations cluster into repeated units: flaky-test prediction [25], defect prediction, program repair.

Data availability. <https://github.com/seanonymoussubmission/agent-monitor-estimand>

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